

## EDUCATION

- **Vellore Institute of Technology** Amaravati, India  
*Integrated M.Tech in Computer Science & Engineering; CGPA: 8.3* 2021 – 2026
- **NRI Junior College** Ongole, India  
*Intermediate (Mathematics, Physics, Chemistry); Percentage: 97.4%* 2021
- **Z.P High School** Rajampalli, India  
*10th Standard; CGPA: 9.7* 2019

## WORK EXPERIENCE

- **AI TASK CONTRIBUTOR (FREELANCE) – PROJECT SHIELD** | GitHub [Soul AI, Apr 2025]
  - Conducted **adversarial prompt testing**, red-teaming, and structured response verification to optimize LLM safety and instruction alignment.
  - Followed rigorous annotation guidelines on a secure platform, executing **Human-in-the-Loop (HITL)** evaluation to benchmark NLP model reliability.

## PROJECTS

- **DEEPSCOPE – AUTONOMOUS MULTI-AGENT DEEP RESEARCH SYSTEM** | GitHub [Feb 2026]
  - Architected an autonomous multi-agent research workflow using **LangChain Deep Agents** and **LangGraph**, coordinating specialized subagents (*Lead, Topic Researcher, Fact Verifier*) to conduct parallel web investigations.
  - Implemented a disk-backed **Filesystem Workspace** to isolate raw findings from LLM context memory (mitigating hallucination) and synthesized verified Markdown research reports.
- **NEXUSRAG – ENTERPRISE-GRADE RAG LIFECYCLE PLATFORM** | GitHub [Jan 2026]
  - Built an end-to-end RAG platform with a **7-stage visual ingestion pipeline**, implementing **Dense + Sparse Hybrid Search** (ChromaDB + BM25 via Reciprocal Rank Fusion) for semantic and keyword retrieval.
  - Integrated **FlashRank cross-encoder reranking** and **Groq LPU inference**, delivering sub-second query latency and grounded answers with verifiable page citations.
- **E-COMMERCE AI ANALYTICS – AUTONOMOUS AGENTIC DATA ANALYST** | GitHub [July 2025]
  - Developed a text-to-SQL analytics agent using **LangGraph StateGraph**, **Groq**, and **Gemini Flash Lite**, featuring a **self-healing retry loop** that auto-diagnoses SQLite errors and corrects SQL.
  - Enforced security via **Meta Prompt Guard 86M** and read-only DB sandboxing, exposing analytics tools via **Streamlit** and **Model Context Protocol (MCP)**.

## TECHNICAL SKILLS

- **Agentic AI & Orchestration:** LangGraph, LangChain, Multi-Agent Systems, Deep Agents, Tool/Function Calling, Self-Healing Loops, Model Context Protocol (MCP), Structured Outputs (Pydantic)
- **Advanced RAG & Retrieval:** Dense + Sparse Hybrid Search, BM25, Reciprocal Rank Fusion (RRF), Cross-Encoder Re-ranking (FlashRank), ChromaDB, Vector Embeddings, Semantic Chunking
- **LLMs & Machine Learning:** Meta LLaMA 3.x, Google Gemini, Groq LPU, Prompt Engineering, Meta Prompt Guard 86M, HITL Evaluation, Hugging Face, PyTorch, Scikit-Learn
- **Engineering Tools & Languages:** Python, Java, SQL, FastAPI, Streamlit, SQLite, Git, GitHub Actions, Docker, RESTful APIs, Data Structures & Algorithms

## CERTIFICATIONS

- **Data Structures and Algorithms** — Take U Forward
- **Data Science, Machine Learning, Deep Learning** — KrishAI Technologies
- **Generative AI** — KrishAI Technologies